SN Computer Science (2025) 6:335
https://doi.org/10.1007/s42979-025-03848-y

ORIGINAL RESEARCH

A Quest for Context-Specific Stock Price Prediction: A Comparison Between Time Series, Machine Learning and Deep Learning Models

Mugdha Shailendra Kulkarni[1] · S. Vijayakumar Bharathi[1] · Arif Perdana[2] · Divisha Kilari[1]

Received: 29 July 2023 / Accepted: 3 March 2025
© The Author(s) 2025

Abstract

Understanding the complexities of buying, selling, and holding stocks is crucial for institutional and individual investors to make informed decisions. Despite their significance, many investors face challenges in this area. Accurate stock price forecasting is a vital tool that aids investors in making profitable decisions. This study evaluates stock trends and patterns with an in-depth analysis of the Bombay Stock Exchange (BSE) stock data. We utilized various techniques, including time-series analysis, machine learning, and deep-learning models. This investigation spanned two distinct datasets: one with and one without COVID-19 stock price data. By comparing the outcomes, we seek to identify the most effective model for stock price prediction. Our findings indicate that each model has its strengths and limitations. Time series models accurately forecast short-term stock prices, whereas machine learning models demonstrate superior generalization capabilities. Deep learning models, however, stand out for their ability to predict long-term stock prices more accurately. Understanding each model's performance nuances is crucial for institutional and individual investors and regulators to optimize their strategies and decision-making processes.

Keywords Stock price prediction · Machine learning · Deep learning · Time-series analysis · ARIMA · LSTM

Abbreviations

EIC	Economic, industry, and company analysis
GDP	Gross domestic product
ARIMA	Auto-regressive integrated moving average
SMA	Simple moving average
SVR	Support vector regression
RNN	Recurrent neural networks
LSTM	Long short-term memory
RFR	Random forests regression
XGBoost	Extreme gradient boosting
ACF	Autocorrelation function
PACF	Partial autocorrelation function
AIC	Akaike information criterion
SBC	Schwartz criterion
BIC	Bayesian information criterion
MLP	Multilayer perceptron
ME	Mean error
RMSE	Root mean square error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MASE	Mean absolute scaled error
ACF1	Autocorrelation of errors at lag 1
RBF	Radial basis function

✉ Arif Perdana
arif.perdana@monash.edu

Mugdha Shailendra Kulkarni
mugdha@scit.edu

S. Vijayakumar Bharathi
svkbharathi@scit.edu

Divisha Kilari
kilari.divisha@associates.scit.edu

[1] Symbiosis Centre for Information Technology, Symbiosis International (Deemed University), Pune, India

[2] Monash University, Monash, Indonesia

Introduction

The stock market, a cornerstone of modern economies, has long captivated researchers and practitioners due to its complexity and far-reaching implications. At its core, stock price fluctuations are governed by fundamental principles of supply and demand [1]. This seemingly simple mechanism is, however, an intricate web of influencing factors that span the economic, industrial, and corporate domains [2–5]. Macroeconomic forces such as inflation rates, unemployment

Published online: 02 April 2025

figures, and gross domestic product have a significant influence on stock prices [6, 7]. However, the sensitivity of the market extends beyond quantifiable metrics. The power of the information, whether positive or negative, cannot be underestimated. Corporate events, from mergers and acquisitions to changes in governance structures, can cause dramatic shifts in share prices [8–10]. Indeed, unexpected negative news can be a particularly potent catalyst for market movements [11].

In this dynamic environment, investors traditionally rely on two primary approaches: technical and fundamental. The human elements, however, introduce additional layers of complexity. Factors such as overconfidence, herd behavior, and irrational reactions to news can significantly influence the decision-making processes [12, 13]. This interplay between rational analysis and market sentiment underscores the challenge of accurate price prediction [14, 15]. The need for reliable stock price forecasting has led to the development of diverse analytical models. These range from traditional statistical methods to cutting-edge machine learning and deep learning approaches. The ultimate measure of any prediction model is its ability to minimize error rates [10, 16, 17]. Conventional forecasting techniques rooted in time-series analysis of historical data have long been the industry standard [18]. Although these methods excel in short-term analyses, their efficacy often wanes when applied to longer time horizons [15, 19].

Data science ushered in a new era in stock market forecasting. With their ability to discern patterns from vast datasets, machine-learning models have gained significant traction [16, 20–22]. In particular, supervised learning algorithms have shown promise in classifying new data based on historical patterns, proving especially useful for medium-to-long-term forecasts [23]. Deep learning algorithms, leveraging the power of neural networks, push the boundaries further by recognizing complex patterns and predicting market sentiment [24–29].

The coexistence of these approaches—traditional, machine learning, and deep learning—has sparked a crucial line of inquiry in the field. Researchers have emphasized the importance of comparative studies in assessing the relative merits of each method [30–32]. These investigations have shed light on the performance characteristics of individual prediction techniques [20, 24, 33]. A critical gap remains, however, in our understanding of the specific context in which each model excels.

Our study aims to identify the most suitable model for context-specific stock price forecasting, whether for short- or long-term predictions or broader generalizations. By offering insights across different periods, this study serves institutional and individual investors and analysts alike. This highlights the optimal performance of various forecasting techniques under specific market conditions, including time-series, machine learning, and deep learning models. For traders focusing on short-term strategies, this study provides a valuable analysis of models such as Simple Moving Average (SMA) and Long Short-Term Memory (LSTM), demonstrating strong performance in handling real-time data. In contrast, investors with a long-term perspective can benefit from the study's insights into models such as Autoregressive Integrated Moving Average (ARIMA) and eXtreme Gradient Boosting (XGBoost), which offer robust predictions over extended periods.

The objectives of this study are threefold: first, to analyze the predictive performance of different methods and determine their precision and dependability in forecasting stock prices; second, to evaluate the effectiveness of these models across short-, medium-, and long-term horizons; and third, to examine the impact of various market conditions on the performance of these forecasting models, particularly in the context of disruptive events such as the COVID-19 pandemic. These objectives coalesce into our central research question: *In which scenarios do time-series, machine learning, and deep learning models demonstrate superior predictive power in stock price forecasting?*

We leverage a decade of historical stock data (2010–2021) from the Bombay Stock Exchange (BSE) to address this question, focusing on closing values and significant market trends. The choice of BSE is deliberate, given its global recognition and high concentration of small-scale companies known for stock price volatility. The impact of the pandemic on these companies, set against the backdrop of the BSE's reflection of emerging market trends, offers a unique lens through which to analyze market behavior and resilience.

Our study contributes to data science and stock price analysis by bridging the gap between theoretical understanding and practical applications in financial markets. We comprehensively compare established stock price forecasting models across different periods, focusing on crisis events, such as the COVID-19 pandemic. This comparative analysis, spanning time series, machine learning, and deep learning models, yields model-specific recommendations that provide investors with nuanced strategies based on different time horizons and market conditions.

Our findings underscore the superiority of deep learning models, particularly the Multilayer Perceptron (MLP), for long-term forecasting and adaptability to market shocks. Concurrently, we highlight the effectiveness of machine learning models such as Random Forest Regression (RFR) and XGBoost for short-term forecasts and trend deviation detection. Our study complements previous research where the impact of extreme market events on model selection and performance has often been overlooked [34–36]. This study emphasizes the need for model diversity and adaptability, demonstrating that no single model exists in all scenarios. We also focus on the trade-off between model accuracy and

interpretability, contributing to the ongoing dialogue on transparency in financial forecasting. By providing specific, actionable insights for institutional and individual investors and regulators, our research offers a solid foundation for informed decision-making in stock price predictions.

This paper is structured as follows: Sect. "Literature review" reviews the models and literature. Sect. "Data and method" outlines the data and methods. Sect. "Analysis and results" analyzes model results. Sect. "Discussion" discusses findings and compares models by error metrics. Sect. "Contributions" explores implications for theory and practice. Finally, Sect. "Conclusion" concludes by highlighting the most effective models, study limitations, and directions for future research.

Literature Review

This section explores prior studies on stock market forecasting, focusing on traditional and advanced predictive models. We describe the use of techniques ranging from simple moving averages to time-series analyses and machine learning and deep learning models. This review categorizes the factors influencing these forecasts into macroeconomic and microeconomic components, offering a structured approach to understanding the complex dynamics. Key research was conducted to highlight the strengths, weaknesses, and developments in the predictive accuracy of these models. This comprehensive analysis lays the foundation for comparative evaluation.

We thoroughly explore stock market forecasting techniques and address several critical issues. First, it emphasizes the importance of stock market forecasting in both academic research and practical applications, setting the stage for our study and underscoring the ongoing need to improve prediction accuracy, highlighted by recent studies. Additionally, the review distinguishes between macroeconomic and microeconomic factors affecting stock prices, offering a nuanced understanding of their various influences on forecasting.

Our review includes a detailed discussion of widely used time-series models, such as ARIMA, SMA, and Holt-Winters, addressing their theoretical foundations, practical applications, and limitations. ARIMA's linearity, for example, suggests a need to explore alternative methods. We also examine machine learning models such as Random Forest, XGBoost, and Support Vector Regression (SVR), highlighting their ability to model complex non-linear relationships in financial data, reflecting their growing importance in the field. Table 1 summarizes the models used to analyze in this study.

Stock market forecasting continues to be the focus of research in academia and industry, particularly concerning two critical aspects: influencing factors and forecasting techniques for improved prediction accuracy [23, 28, 37, 38]. Forecasting factors are divided into macroeconomic elements (major market drivers, such as gross domestic product, monetary policy, inflation, and interest rates) and microeconomic elements (minor market drivers, such as random events, market rumors, and investor sentiment). Financial data, especially from stock indices, are time-series data influenced by various dynamic factors and are crucial for predicting stock prices [19, 39, 40].

Time-series analysis is the most widely used model in statistics-based forecasting techniques [41, 42]. For example, the ARIMA model uses time-series data. However, selecting ARIMA for prediction may not always lead to optimal results because its prediction pattern is directional [41]. A major limitation of this model is its linearity, which limits its ability to capture non-linear trends [43]. For the comparative analysis in this study, we used the SMA, ARIMA, and Holt-Winters tests as representative models for time-series analysis to provide a comprehensive overview of the capabilities and limitations of these techniques [3, 44, 45].

Table 1 Techniques and models in stock market prediction

Aspect	Details	References
Time-Series Models	**Common Techniques:** ARIMA, SMA, Holt-Winters **Key Points**:—ARIMA is widely used but has limitations due to its linearity, making it less effective for non-linear trends.—SMA and Holt-Winters are also evaluated in this study for a comprehensive overview	[3, 41–45]
Machine Learning Models	**Selected Models:** Random Forest, XGBoost, SVR **Key Points**: These models are used to evaluate their effectiveness in predicting stock prices	[5, 46, 47]
Deep Learning Models	**Selected Models:** LSTM and MLP **Key Points**: LSTM, a Recurrent Neural Network (RNN) type, excels in time-series forecasting by effectively managing long-term dependencies through selective data retention or discarding during processing. On the other hand, MLP Neural Networks, composed of multiple layers, are primarily used for classification tasks and learning complex non-linear relationships in data. Together, they provide powerful tools for handling diverse predictive modelling challenges, from sequence processing to classification	[17, 28, 37, 38, 48–51]

Additionally, to evaluate the capabilities of machine learning models in predicting stock prices, we used three different models: random forest, XGBoost, and Support Vector Regression (SVR) [5, 46, 47]. LSTM, a type of RNN, is characterized by its ability to determine when to discard data from memory during processing [28, 37, 38, 48]. In contrast, MLP neural networks are primarily used to solve classification problems [17, 37, 49–51]. The effectiveness of these models was evaluated based on their residual error. The following sections discuss conventional time-series, machine learning, and deep learning models in more detail to provide a comprehensive overview of their respective strengths and limitations in stock price predictions.

Time-Series Models

Time-series models are widely used to assess historical trends and patterns in stock market data [52]. This analytical process systematically collects data points for one or more variables across different time intervals [39]. Notably, these market data are often used to establish the time momentum, suggesting that the predictive value of a stock is intrinsically linked to its historical performance [53, 54]. The utility of time-series analysis extends beyond mere data collection; it plays a crucial role in monitoring, analyzing, predicting, and controlling data streams [43, 55, 56]. In the context of market forecasting, this approach is particularly relevant given that it often hinges on the interrelationships within time-series data. Central to understanding time-series data are three fundamental elements: first, the concept of stationarity, where the mean and variance are consistent over time, and the covariance remains time-independent; second, seasonality, which pertains to predictable and cyclical variations in data; and third, autocorrelation, which refers to the dependencies that exist between variables within a specified time interval. These elements underscore the intricate nature of time-series data and their significance in financial analyses and forecasting [57].

Simple Moving Average

SMA is a fundamental tool in the financial analysis domain, particularly in the context of stock price predictions. Its widespread application and efficacy have been well-documented in the literature [44, 58, 59]. The SMA serves as a pivotal model in forecasting and a crucial instrument in evaluating technical indicators within financial markets [59]. The effectiveness of the SMA model is often validated through various statistical parameters such as the mean absolute percentage error (MAPE), mean absolute deviation (MAD), and mean square error (MSE). These metrics play a critical role in confirming the reliability and accuracy of SMA as a predictive tool [45, 60]. The operational mechanism of SMA involves aggregating closing stock prices over a set period. The average is computed by dividing the aggregate of prices by the number of periods, smoothing out short-term fluctuations and providing a clear view of the long-term trend.

A key aspect of SMA's utility is its responsiveness to market dynamics [61]. Short-term averages are noted for their ability to reflect changes in stock prices and offer timely insight into market movements. By contrast, long-term averages tend to respond more slowly, which can be advantageous in identifying more sustained trends and minimizing the impact of short-term volatility. The ability of SMA to signal trends is particularly noteworthy. For instance, an upward trend in SMA is often indicative of a potential increase in stock prices. This characteristic makes the SMA an invaluable tool for investors and analysts, aiding the strategic decision-making process by providing a more robust understanding of market trends and price movements [62, 63].

Auto-Regressive Integrated Moving Average (ARIMA)

ARIMA, a statistical model noted for its effectiveness in time-series forecasting, operates primarily on non-random structures within data, rather than purely random distributions. This model employs autoregressive techniques to utilize past data points for predictions and integrates its previous values in a regression-like manner to enhance forecasting accuracy [31, 41, 64]. In stock market forecasting, ARIMA considers stock prices as time-lagged variables [41, 65]. However, while ARIMA's autoregressive capabilities are helpful, relying solely on historical stock prices can be limited by external market influence.

A pivotal step in ARIMA modeling is transforming the data into a stationary state, typically through differencing. Stationarity is essential for ARIMA and is generally assessed using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) along with the p-value [65, 66]. The p-value is instrumental in determining stationarity, but the 0.05 threshold is not absolute and may vary based on context and data characteristics. If the p-value is above this threshold, it suggests non-stationarity, indicating the need for differencing. Once stationarity is achieved, the ACF and PACF plots guide the model specifications [31, 56, 65, 66]. Model adequacy was evaluated using the Akaike Information Criterion (AIC) and Schwartz Criterion (SBC), also known as the Bayesian Information Criterion (BIC), with lower values indicating a better fit [65, 66]. It is essential to consider additional model selection criteria based on the specifics of the analysis.

Holt-Winters Method

The Holt-Winters forecasting method, recognized for modeling seasonality and trends in time series, is adept at interpreting patterns such as demand fluctuations [67–69]. Tiwari et al. [70] compared their model with ARIMA and noted its utility in forecasting stock prices. Its strength in identifying data anomalies benefits trend and seasonal variation analysis [69–71]. Gelper et al. [72] emphasized its relevance in business contexts with evolving trends and seasonal correlations. Comprising level, trend, and seasonality components, known as triple exponential smoothing Holt-Winters, provides a nuanced understanding of time-series data [67–69]. The level is adjusted for the local mean, the trend identifies data patterns, and seasonality captures the cyclical patterns. This integration allows adaptation to dynamic data, making Holt-Winters a sophisticated solution in various domains, especially in seasonal business forecasting. While Holt-Winters is effective, it is crucial to recognize that it is not universally superior to other methods, such as ARIMA; the choice depends on the specific data context [67–70]. It can encounter difficulties with complex seasonal patterns, outliers, and structural changes in data. Accurate forecasting requires careful selection of the smoothing parameters. Like all methods, Holt-Winters has limitations and cannot guarantee perfect predictions, emphasizing the need for careful application and consideration of its suitability for a specific forecasting scenario.

Machine Learning Models

Machine learning input is divided into training and testing datasets [69, 73]. It predicts the test data based on rules obtained by learning from the training dataset. In contrast, an ensemble combines various methods to achieve optimal results [74]. This study used three machine learning models–SVR, RFR, and XGBoost regression–to analyze and predict stock prices on a given dataset. Regression is a supervised machine learning technique for both continuous and numerical outputs. Regression models are typically used for problems involving "how much" or "how many."

Support Vector Regression (SVR)

SVR, a supervised regression algorithm that supports linear and non-linear regressions, is suitable for continuous variable predictions such as stock prices [75–77]. SVR is derived from SVM and differentiated by focusing on continuous rather than categorical outputs [46, 75]. In SVR, a hyperplane in a multidimensional space is identified, which aims to maximize the margin between the closest data points, thereby enhancing prediction accuracy [76]. SVR has been noted for its effective generalization performance in stock price predictions [78, 79]. However, it is important to contextualize these benefits. Although SVR can offer robustness against individual observation errors and superior scalability, its performance is relative and may vary compared to other regression models. The mechanism of SVR, involving the transformation of input data into separate hyperplanes, lacks a detailed discussion of key aspects, such as kernel functions and parameter tuning, which are crucial for its application in finance [46, 75, 76, 78].

Moreover, SVR's challenges, such as its sensitivity to data quality and feature selection, have not been addressed [75, 77, 78]. These factors are particularly relevant in financial contexts, where interpretability and model transparency are essential. Although empirical evidence supports the use of SVR in stock price prediction [75–77], a more comprehensive comparison with other methods and including specific results would provide a more balanced and substantiated argument.

Random Forests Regression (RFR)

Random forest is an ensemble method that makes predictions by averaging the predictions of several independent base models called decision trees. This is a general-purpose classification and regression method [80]. Decision trees in random forests are technically independent. Each tree learns from a different bootstrap sample, reduces correlations, and improves the robustness. In particular, RFR divides the training dataset based on the best decision based on the impurity measure. Each dataset was assigned to a decision tree that processed and provided an output. This RFR compares the predictions from all trees and then combines the projections to arrive at a single, generally averaged result [4, 47]. RFR is an ensemble of decision trees used in regression problems [81]. It uses bootstrap aggregation or bagging to reduce variance in statistical learning methods [82]. While constructing a decision tree, bootstrap samples of size B are generated from size N data by randomly drawing observations, which significantly reduces variance [77]. The average of the multiple estimates can be calculated for M different trees to compute the ensemble. The data were then split recursively into separate parts. The decision tree algorithm iteratively creates roots, leaves, and segments of leaf nodes [38, 83]. For a node, data partitioning is based on a criterion for an impurity measure, such as Shannon entropy or Gini impurity. The decision to split is based on the highest information gain or impurity reduction [4]. A node with the lowest impurity index is considered a leaf node and is not classified further.

eXtreme Gradient Boosting (XGBoost) Regression

XGBoost, akin to gradient boosting, applies boosting techniques to weak learners who are typically shallow decision trees. It stands out from traditional gradient boosting methods because of its enhanced features for cross-validation, regularization, pruning, and more effective avoidance of overfitting [27, 84]. XGBoost does not train decision trees based solely on the errors of previously created trees; it utilizes the gradients of these errors, offering more directional insight for refining the model [64, 84]. The process does not begin with a predefined regression predictor but with an initial prediction, often the average of the target variable. From this starting point, XGBoost iteratively enhances the prediction by adding new trees, explicitly focusing on the gradients of errors from previous iterations [84]. This emphasis on "weak learners or simple decision trees with limited depth aids in mitigating overfitting while enhancing flexibility. Rather than applying gradient descent to each tree, XGBoost employs an approximate second-order optimization technique to efficiently determine the best split in each tree. This sophisticated approach enables constructing an optimal ensemble model through successive improvements [27, 64, 84].

Deep Learning Models

A neural network performs a better approximation and can better map the relationship between input and output data for complex datasets [37]. It also efficiently identifies new samples not used to train the dataset [21]. Neural networks consist of input, hidden, and output layers, respectively. The input layer comprises artificial neurons that bring data into hidden layers. The hidden layers then determine the nodes by weighting inputs and producing an activation function. The output layer is responsible for generating the final result. The synapses connecting the nodes carry a weight used for decision-making in the output layer. Backward propagation was then used to determine the optimal error value [17].

Long Short-Term Memory

In financial market analysis, LSTM is a pivotal tool adept at deciphering the complexities of stock price movements. Siami-Namini and Siami-Namin [31] underscored the unique gated structure of LSTMs, in which intelligent filters governed by sigmoid functions adeptly retain significant historical data and discard inconsequential details. This selective memory is crucial in stock market analysis, where discerning long-term trends from market noise can be essential to predictive accuracy. LSTM has been recognized for its ability to capture temporal dependencies, which are key aspects in predicting stock market movements [26, 48, 85, 86].

The application of deep learning in stock market prediction has been extensively researched, with LSTMs playing a central role. Prior research illustrates that LSTMs capture the complexity of financial data relationships [21, 26, 29, 33, 87]. These findings offer invaluable insights, aiding investors and analysts in navigating the intricate web of market dynamics. An essential feature of LSTMs is their resilience to overfitting, a notorious challenge in deep-learning models [88, 89]. The gated mechanism and focus on long-term dependencies equip LSTMs to generalize more effectively to new data, thereby ensuring more stable and trustworthy forecasts in the unpredictable realm of the stock market [86].

Multilayer Perceptron Regressor

MLP is a neural network-based model for effectively handling non-linear data [51, 90]. Central to its operation is the backpropagation algorithm, which employs an activation function to navigate through the network's multiple layers of neurons [91, 92]. In the context of stock price prediction, a regression task, the MLP initially assigns random weights to perceptrons. The model then uses a forward computation process across input, hidden, and output layers to calculate predicted stock prices. Unlike traditional applications, where classification is a primary function, MLP focuses on predicting a continuous value and the future price. The accuracy of these predictions was crucial, and the error rate was computed to gauge this accuracy. The error, representing the difference between the predicted and actual prices, was calculated using methods such as mean squared error, which is particularly relevant for regression tasks [91, 92]. Table 2 provides an overview of forecasting models and their core features.

Data and Method

The prediction generally follows a series of steps to achieve accurate results. Regardless of the programming language or model used, the steps for obtaining the predicted values are always the same [93]. Data are essential for building and training the predictive models. Therefore, data collection and processing are crucial for model development [94]. Data collected from the Internet or organizations may be unprocessed and require further cleaning. Data must first be collected from valid sources for prediction using selected models. The collected data may also contain impurities, such as missing elements or noisy data. Therefore, data must be cleaned and converted into a suitable format. Once data cleaning was complete, the data were divided into training and testing data. The training dataset was used to train the predictive models. The test data were used to evaluate the models. Once the values are available, each model can be

Table 2 Forecasting Models and Their Key Aspects: Selection Rationale and Strengths

Category	Model	Rationale for Selection	Key Strengths	Citations
Panel A				
Time-Series Models	**Simple Moving Average (SMA)**	Selected for its reliability in identifying trends while filtering out short-term market noise	Effective in smoothing fluctuations and revealing long-term trends	[19, 39, 40, 44, 58, 59]
	ARIMA	Chosen for its ability to handle non-stationary data through differencing, crucial for typical stock price patterns	Captures temporal dependencies effectively for short-term predictions	[19, 39–41, 65]
	Holt-Winters	Included for its capability to decompose time series into level, trend, and seasonal components for nuanced analysis	Effective in identifying seasonal and trend-based variations in financial data	[19, 39, 40, 67–69]
Machine Learning Models	**Support Vector Regression (SVR)**	Selected for its effectiveness in handling continuous variable predictions and robust generalization performance	Excels in capturing non-linear relationships in stock price data	[75–77]
	Random Forest Regression (RFR)	Chosen for its ability to reduce variance through bootstrap aggregation and handle high-dimensional financial data effectively	Provides robust generalization and reduces overfitting	[4, 47, 80]
	XGBoost	Included due to its superior performance in gradient boosting, enhanced regularization, and cross-validation features	Prevents overfitting and adapts well to volatile financial markets	[27, 84]
Deep Learning Models	**Long Short-Term Memory (LSTM)**	Chosen for its unique gated structure, enabling it to maintain long-term memory and discern long-term stock price trends	Captures temporal dependencies and adapts well to complex time-series data	[26, 48, 85, 86]
	Multilayer Perceptron (MLP)	Selected for its ability to model non-linear data and learn complex relationships through multiple layers of neurons	Provides high accuracy and robust handling of complex, non-linear stock price relationships	[37, 38, 51, 90, 91]
Panel B				
Key Aspects	**Model**	**Rationale for Selection**	**Key Strengths**	**Citations**
Temporal Dynamics	All Time-Series Models	Focus on capturing historical patterns and seasonality in stock price data	Provide insights into short-term trends based on historical performance	[52, 53]
Non-linear Relationships	All Machine Learning Models	Excel in identifying and modeling complex, non-linear relationships within stock price data	Enable robust pattern recognition and generalization	[73, 74]
Long-term Dependencies	LSTM, MLP	Effective in modeling long-term dependencies and intricate patterns in financial data	Adapt well to volatile and complex financial market scenarios	[26, 37, 38, 48, 85, 86]
Generalization Capability	RFR, XGBoost	Ensemble methods robustly generalize unseen data, ensuring reliability in diverse market conditions	Handle varying data conditions with reduced overfitting	[80, 84]
Adaptability to Market Conditions	All Models	Comparative analysis across different market conditions, including the COVID-19 period, provides comprehensive understanding of strengths and limitations	Enables evaluation of trade-offs between model complexity and practical applicability in trading scenarios	[30–32]

compared efficiently using specific error measures. The error measures the deviation of the predicted values from the test set. The model with the slightest deviation was the most efficient. The following sections explain how these steps were applied in our research. Figure 1 shows the steps used to develop the predictive models.

Data Collection

The historical stock data were obtained from the BSE website. The data were collected daily and monthly. Data were collected over the last ten years (2010–2021). Figure 2 shows the classification of the training and test data. We considered the closing value and the strong upward and downward movements in the time-series data. We investigated the generated charts to identify patterns. We used R programming to implement the time-series models.

In contrast, Python was used to run the machine learning and deep learning models. An Augmented Dickey-Fuller test was used to determine whether the data were stationary. Failure to reject the null hypothesis indicates a non-stationary data series, which indicates a linear data series (difference stationery). We can also reject the null hypothesis if the value of the test statistic is less than the critical value (i.e., the data are stationary). Alternatively, the data are non-stationary if the null hypothesis is not rejected. The null and alternative hypotheses are defined as follows: the series has a unit root (value of $a = 1$) (Null Hypothesis), and the series has no unit root (alternative hypothesis).

Data Cleaning and Partitioning

Our analysis uses monthly stock price data. We assessed the data for inconsistencies such as missing values or incorrect format of the period closing price. We argue that these data are sufficient for machines to learn and detect patterns in stock-price movements. For further analysis, the data were converted to logarithmic form. The data used for analysis were divided into training and testing data. The engine uses training data to detect patterns in stock price movements. A large dataset facilitates better generalization [80, 95]. Data from January 2011 to December 2018 were used for training, whereas test data were from January 2019 to February 2021. Figure 5 shows the training and testing data. The part highlighted in blue was used for training the models, and the part highlighted in green represented the test data, the data to which the prediction models were applied. Prediction models were applied to the two datasets. The first dataset excludes the COVID-19 stock price data (January 04, 2010, to January 31, 2019). The second dataset includes COVID-19 stock price data (January 04, 2010, to April 15, 2020).

Measures Used to Assess Models' Performance

The error criterion was used to evaluate the efficiency of the model. The difference between the actual and predicted values was used to determine the model performance. Studies have shown that different error criteria focus on various aspects of error characteristics. Therefore, a combination of error criteria helps better evaluate the model [96]. The measures used were mean error (ME), root mean square error (RMSE), mean absolute error (MAE), MAPE, mean absolute scaled error (MASE), and autocorrelation of errors at lag 1 (ACF1).

Analysis and Results

This section presents the parameters defined when running the models and the resulting plots that can be used to analyze the models. The results are for the 10-year dataset, including stock prices during the pandemic.

Results of Time Series Models

Results of Simple Moving Average

This model determines the average price from a selected subset of closing-price data. SMA considers three to ten years to reduce noise and reveal trends/patterns in the data. The resulting chart shows the trends of the values predicted from the test data (Fig. 3). This trend indicates either an upward or downward trend based on which an investment decision can be made. Table 3 summarizes the SMA forecast results from 2010 to 2018.

The assessment of the model's performance, as shown in Fig. 3, offers insights into its predictive abilities. The blue line in the figure represents the forecasted values derived from the SMA model, which may be helpful in contexts characterized by market uncertainty and potential volatility. The magnitude of the residuals or the differences between the actual and predicted values indicates the model's prediction error. Notably, the pronounced peak in residuals around 2016 suggests a significant deviation from actual stock prices, indicating a substantial forecast error. It is essential to emphasize that the ability of the SMA model to track values annually should not be mistaken for high accuracy without supporting quantitative metrics.

In practical applications, the SMA model is generally more effective for short-term forecasting, owing to its inherent simplicity and limitations in capturing complex data patterns. The model's performance within the observed range does not inherently validate its efficacy for long-term forecasting because it may not account for more intricate market



Fig. 1 Steps in developing predictive models



Fig. 2 Original data plotting of dataset

dynamics. Caution is required when extrapolating beyond the established data range. For applications that extend over the long term, it is imperative to assess the responsiveness of a model to new information and its capacity to adapt to evolving market conditions. Such an assessment is critical to ensure that the model remains relevant and accurate in a landscape where many unpredictable factors can affect financial time series.

Results of ARIMA

ARIMA is used when data are non-stationary and exhibit a trend over time. We first used the Dickey-Fuller test to determine whether the data met the assumption of stationarity. The test yielded a significance value of 0.5478 (>0.05), indicating that the data were non-stationary. In the ARIMA model, we convert time-series data of a stock market into stationary data using differencing. Differencing is a technique used to convert non-stationary data into stationary data. The built-in *auto.arima* function was then used to determine the required parameters. We used the AIC to measure the goodness of fit. This helps to choose simpler models over complex ones to avoid overfitting. The ACF plot shows the autocorrelation of lagged values. The PACF plot shows the correlation between the residual values (Fig. 4). Figure 5 presents the ARIMA forecast's residual plot, and Table 4 summarizes the ACF and PACF plot lags.

The ACF plot in Fig. 4 underscores the inherent autocorrelation at lag 0, which is an expected feature indicative of a time-series correlation with its current value. A near-unit value at this juncture signals robust autocorrelation, indicating the significant influence of preceding values on future outcomes. Nevertheless, the accurate measure of correlation and potential predictive power lies in the significance of the ACF values in the subsequent lags. Here, values within the confidence interval bounds, as depicted by the shaded area, denote statistical insignificance, challenging any presumed relationship at those points. The PACF plot further complements this analysis by isolating the correlation at each lag after accounting for the correlations at all shorter lags, aiding in discerning autoregressive model order. However, contrary to the

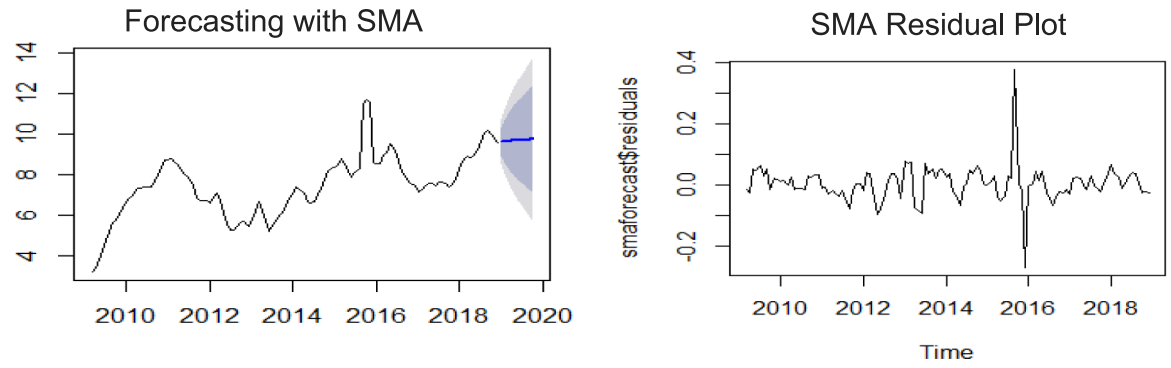


Fig. 3 Forecasting with SMA and its residual plot

Table 3 SMA forecast results

Year	Value	Residual Plot
2011	6	0
2012	7	0
2014	7.50	0.08
2016	8	0
2018	8	0.10

predictive implications, PACF is a diagnostic tool, not a predictor.

The apparent randomness and lack of pattern in the residual plot suggest the model's efficacy in capturing the essence of the data, leaving the residuals as mere noise, indicative of a well-fitted model. However, a comprehensive evaluation requires juxtaposition against the ACF plot of the original data to ensure that all correlations are adequate. These figures seem to account for stock price fluctuations, with the residual plot affirming data absorption within the model structure. Nonetheless, a balanced interpretation necessitates acknowledging the diagnostic role of the ACF and PACF plots, whereas the randomness of the residual plot must be scrutinized to confirm the model's ability to reflect market volatility and uncertainty. This careful consideration of the diagnostic plots holistically confirms the model's validity and potential refinements.

Results of the Holt-Winters Model

The Holt-Winters method typically applies three levels of the triple exponential first, second, and third seasonal components. Three order parameters are considered: Alpha, Beta, and Gamma. Alpha is the coefficient of level-smoothing. Beta is the coefficient for trend smoothing, and gamma specifies the coefficient for seasonal smoothing. As no clear trend or seasonality was observed in the data, we used a simple exponential smoothing method for the prediction. Model < -HoltWinters (training data, beta = FALSE, gamma = FALSE). Thus, the level through alpha, slope estimates the trend through beta, and seasonality components through gamma. The range for the three-parameter controls was between 0 and 1, indicating relative weight. Here, except for alpha, we considered both parameters false by maintaining alpha to control for smoothing. When the alpha is closer to 0, recent data are weighted more than past data. Figure 6 and Table 5 present the forecast results using the Holt-Winters model.

Based on Fig. 6 and Table 5, the output of HoltWinters () shows that the estimated value of the beta parameter was

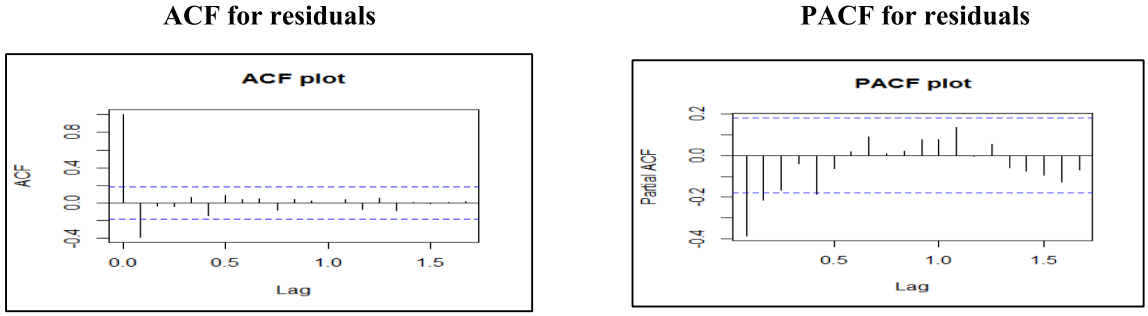


Fig. 4 ACF residual plot and PACF residual plot

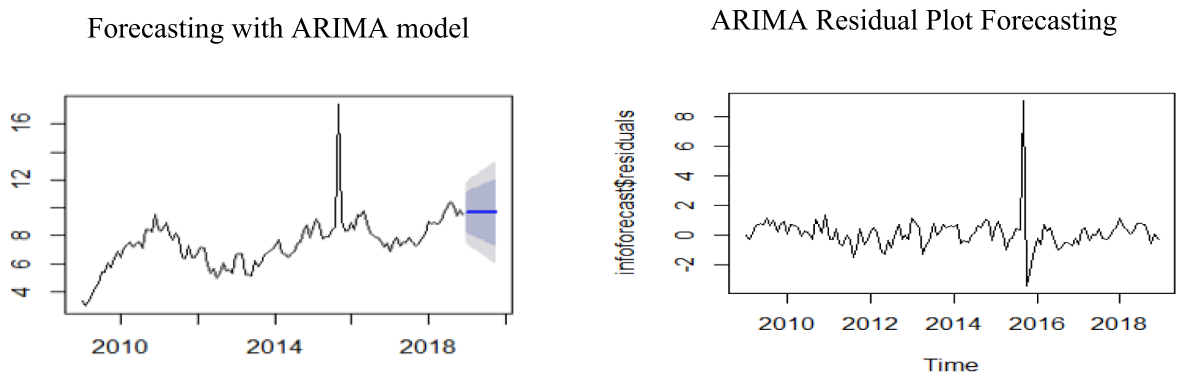


Fig. 5 Forecasting with ARIMA and its residual plot

Table 4 Lags in ACF and PACF plots

Lag	ACF plot	PACF plot
0.5	0.10	−0.05
1	0	0.10
1.5	−0.50	−0.10

0.80. When approached 1.0, the most recent data values had higher weights. Gamma is 0.03, close to zero, meaning that the most recent values have a higher weight. Gamma is set to False because the seasonal component is omitted. The model assessment presented in Fig. 6, which shows a beta of 0.80, indicated that short-term fluctuations were captured. In addition, the gamma value is 0, which shows that the seasonality in this model is not captured. The model emphasizes short-term trends, with a beta parameter of 0.80. As a result, rather than detecting long-term patterns, the Holt-Winters model, as it is now built is better suited for catching and forecasting short-term variations in stock values. The beta parameter affects how easily the Holt-Winters model can adjust to the shifting circumstances. A model with a greater beta value is more sensitive to recent changes in data, and may be better able to adjust to changing market conditions.

Results of Machine Learning Models

Results of Support Vector Regression (SVR)

We used the following parameters for SVR (C = 1000.0, degree = 3, gamma = 0.1, kernel = 'rbf).' where C is the regularization parameter of the error term. We used a radial basis function (RBF) kernel for regression. The degree of the polynomial function is 3. where gamma is the kernel coefficient. The C and gamma values determined the resolution of the hyperparameters of the heat map. An outlier is observed in the SVR RBF kernel output (Fig. 7). Table 6 presents the performance summary of SVR. The results obtained from the two datasets were then presented. Figures 8 and 9 show the *before* and *after* COVID-19 predictions.

According to the model (Fig. 9), the COVID-19 period saw a notable increase in the RMSE, MAE, and MAPE, suggesting that the SVR model's prediction performance declined because of the pandemic's exceptional market conditions. The exceptional volatility and unpredictability of the stock market during the COVID-19 period could cause the outliers observed in the SVR RBF kernel outputs. For the model to be more robust and refined, it is essential to recognize and comprehend these outliers. The selection of SVR with an RBF kernel and particular parameter values suggests an effort to model intricate non-linear correlations in stock-price changes. Disruptive market conditions, however, linked to the COVID-19 crisis may have strained the model's performance.

Results of Random Forest Regression (RFR)

Table 7 presents the results of the RFR analysis. The prediction errors, RMSE and MAE, have increased from 14.5 to 19.57 and 9.68 to 13.15 respectively. The short-term increases in RMSE and MAE indicate that the RFR model could not adjust promptly to the abrupt and dramatic shifts in the market caused by the COVID-19 pandemic, as shown in Figs. 10 and 11. The increased prediction errors indicate that it was difficult to foresee sharp short-term swings and rapid changes in stock prices during volatile times. With these parameters, the RFR model can accurately identify the underlying patterns and trends in the stock market. The comparatively stable MAPE of the model suggests that it consistently maintains a level of percentage accuracy, which can be helpful for long-term investors who prioritize overall trends over cyclical variations.

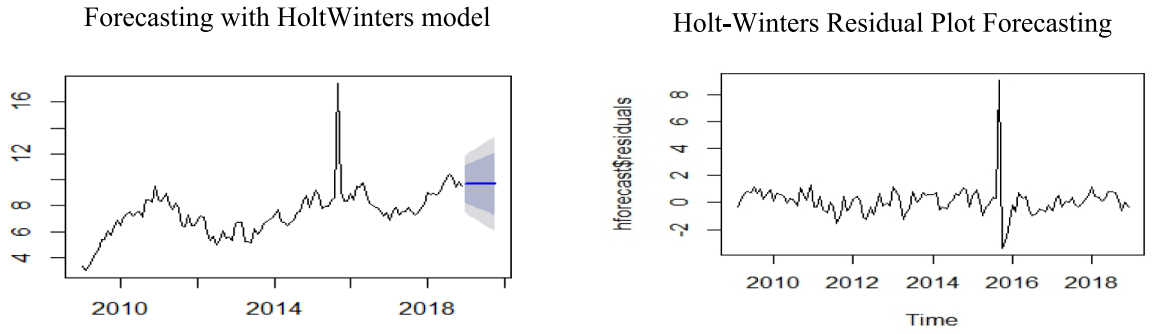


Fig. 6 Forecasting using Holt-Winters and its residual plot

Table 5 Holt-winters forecast results

Time	Forecast	Forecast residuals
2010	4	1
2012	6	0.50
2014	8	1
2016	9	0.80
2018	10	1.20

Results of XGBoost Regression

An XGBoost regression consists of 1000 estimators, and the squared error is chosen as the loss function. Table 8 presents the XGBoost regression results for the pre-COVID-19 stock price data and COVID-19 inclusive datasets. As shown in Figs. 12 and 13, XGBoost regression can predict the trend in both datasets, and MAPE shows a mirror-image increase in the Covid-inclusive datasets, allowing long-term prediction.

According to Fig. 12, the squared error loss function indicates the model's sensitivity to higher errors, which may account for the increased RMSE and MAE during the crisis. The comparatively low MAPE values for both the pre-COVID-19 and full datasets show that, on average, the model maintained a consistent level of percentage accuracy, even in the face of difficulties with short-term predictions. Long-term investors who are less impacted by transient swings and are more focused on broad patterns may find this advantage. The XGBoost regression model can identify the underlying patterns and trends based on its ability to forecast trends in both datasets, as shown in Figs. 12 and 13. Although it had trouble making accurate short-term forecasts during the COVID-19 pandemic, its accuracy remained steady on a percentage basis. Therefore, long-term investors who value broad trends and patterns in stock prices may find this a helpful tool. Research on model parameters and improvements can increase their usefulness in responding to dynamics.

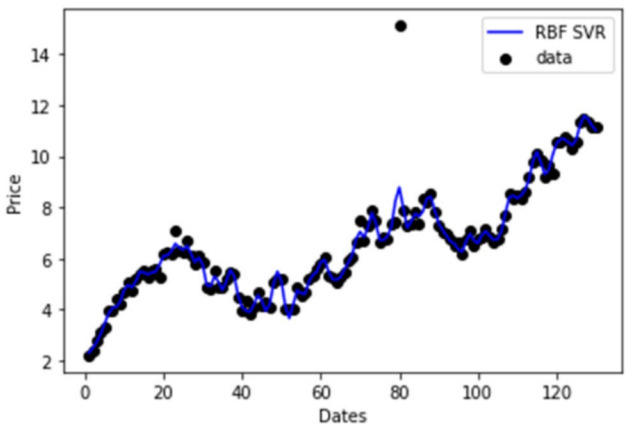


Fig. 7 Outlier in SVR RBF kernel output

Table 6 SVR performance metrics

Performance metrics	Before COVID-19 stock dataset	Before and during COVID-19 stock dataset
RMSE	86.34	249.70
MAE	63.75	164.14
MAPE	8.85	17.48

Results of Deep Learning Models

Results of Long Short-Term Memory

We used a sequential LSTM model. It uses a dense layer architecture to specify the 25 units in the dense () function. After fitting the training data, the loss function returns to 0.0228. The graph shows the predicted values compared to the test data (see Fig. 14).

An assessment of the LSTM model, shown in Fig. 14, is presented in this section. The difference between the expected and actual values was measured using a loss

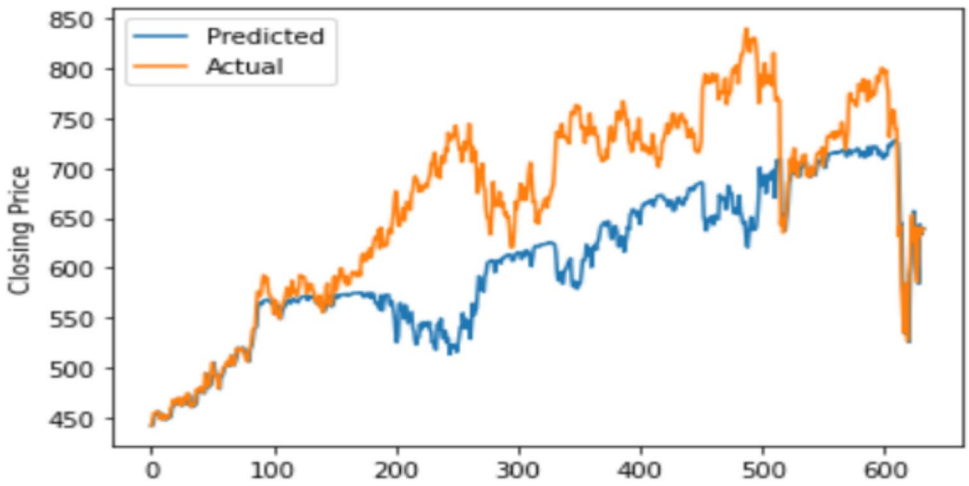


Fig. 8 SVR Analysis (Before COVID-19 Stock Dataset)

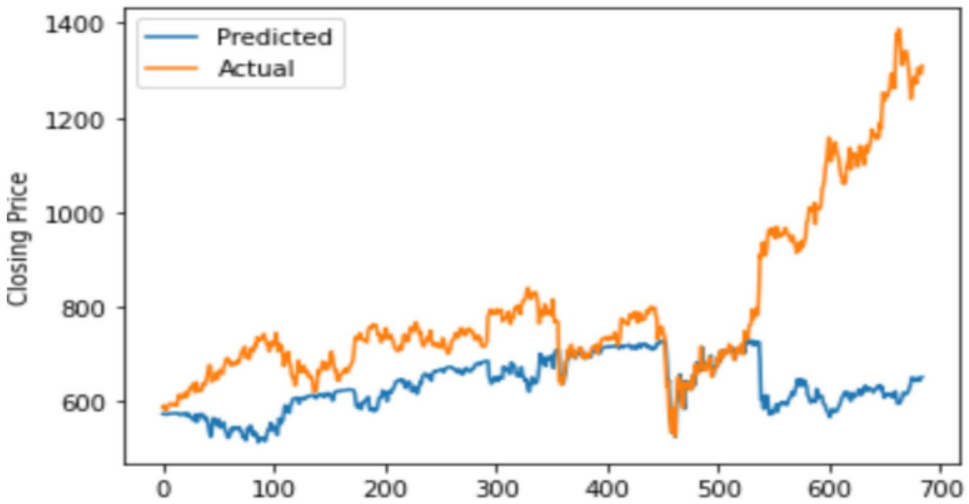


Fig. 9 SVR Analysis (Before and During COVID-19 Stock Dataset)

Table 7 The RFR Performance Metrics

Performance metrics	Before COVID-19 stock dataset	Before and during COVID-19 Stock Dataset
RMSE	14.50	19.57
MAE	9.68	13.15
MAPE	1.45	1.66

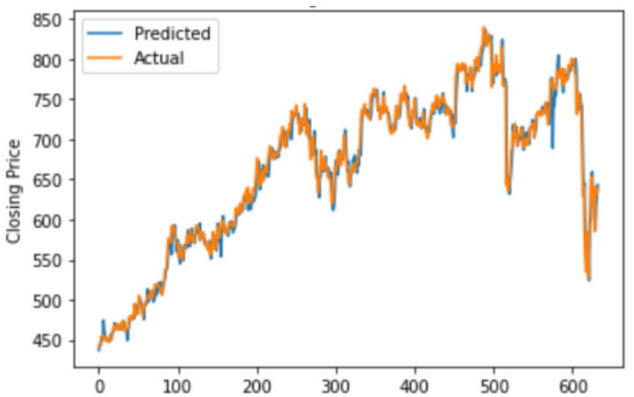


Fig. 10 RFR Analysis (Before COVID-19 Stock Dataset)

function during training. A smaller loss number (0.0228) indicated a better model fit to the training set. The sequential LSTM model can identify short-term patterns and dependencies in the stock market data during training, as indicated by the low loss value (0.0228). This suggests that the training data fits the data relatively well. The long-term efficacy of the sequential LSTM model depends on its capacity to identify underlying trends and patterns that endure over time, although its short-term performance remains significant. The ability of the model to understand complicated interactions was aided by the dense layer consisting of 25 units. This could enable the model to recognize and forecast long-term stock market patterns.

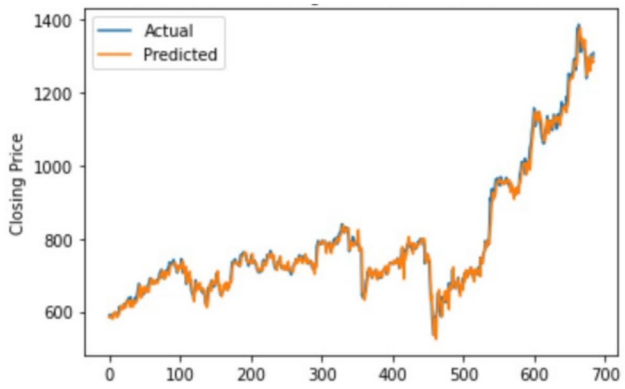


Fig. 11 RFR Analysis (Before and During COVID-19 Stock Dataset)

Table 8 The XGBoost regression performance metrics

Performance metrics	Pre-COVID-19 prediction	Full data
RMSE	15.75	18.58
MAE	10.26	12.65
MAPE	1.53	1.602

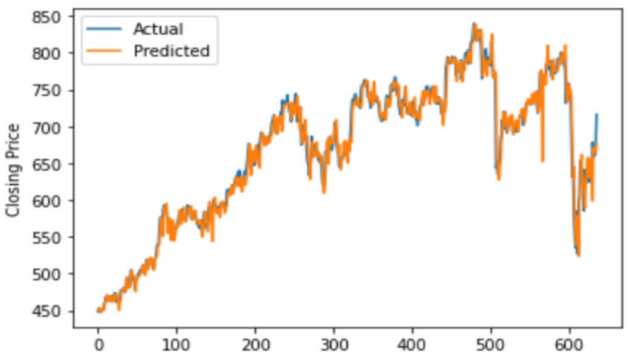


Fig. 12 XGBoost Regression Analysis (Before COVID-19 Stock Dataset)

Results of Multilayer Perceptron Regressor

We used the following data for MLP (hidden_layer_sizes = (13,13,13), max_iter = 500), which followed the same steps as the other prediction models: reading, preprocessing, training, and fitting the data. '13' here indicates the number of neurons in each layer. It uses a rectified linear unit function and returns f(x) = max (0, x). The alpha and beta rates were used as default values. They specify the decay rates of the moment vectors. Table 9 and Fig. 15 present the results of the MLP analysis.

In the model assessment of MLP, investors seek prompt and precise signals, and accuracy can be important in the short term. Positive predictions have a better chance of being correct with a higher precision, which is important for short-term trading decisions. A balance with recall is crucial to prevent the loss of chance, especially in volatile short-term market conditions. In the long term, a higher recall becomes more critical when investors adopt a strategic approach. This guarantees that a greater proportion of equities with long-term growth potential is included in the model. While accuracy is crucial for preventing false positives, investors can create a more extensive long-term portfolio with a higher recall rate.

Discussion

Our analysis revealed apparent differences in forecasting models performed during the pre-COVID-19 and COVID-19 periods, emphasizing the significant impact of changing market conditions on predictive outcomes. The COVID-19 pandemic brought unprecedented volatility, disrupting established stock price patterns and trends. This created challenges for many models, as seen in increased error metrics like RMSE and MAPE during the pandemic. Understanding these differences provides critical insights into model efficacy under varying market conditions. The pre-COVID and COVID-19 periods differed fundamentally, significantly influencing model performance, as explained in the following Table 10.

The comparison between pre-COVID-19 and COVID-19 market conditions highlights the significant challenges models face when adapting to extreme volatility, irregular trading volumes, and unpredictable market sentiment. Market conditions were relatively stable during the pre-COVID-19 period, allowing models to leverage predictable patterns. In contrast, the COVID-19 period introduced sharp price fluctuations, news-driven sentiment shifts, and sporadic trading volumes, disrupting traditional models based on historical data. Models relying on consistent patterns struggled, while more flexible machine learning and deep learning approaches showed varying degrees of resilience, with deep learning models demonstrating superior stability under market turbulence. Table 11 details each forecasting models performance.

Table 11 provides a comprehensive comparative analysis of the performance of various stock price prediction models during the pre-COVID-19 and COVID-19 periods. The models are categorized into three main types: time-series, machine-learning, and deep-learning. Each model's performance is assessed using key metrics such as RMSE, MAE, and MAPE, offering insights into their adaptability and robustness under varying market conditions.

Time-series models (ARIMA, SMA, and Holt-Winters) demonstrate their reliance on stable historical patterns. ARIMA exhibited more significant absolute errors (RMSE) during COVID-19 but improved relative accuracy (MAPE),

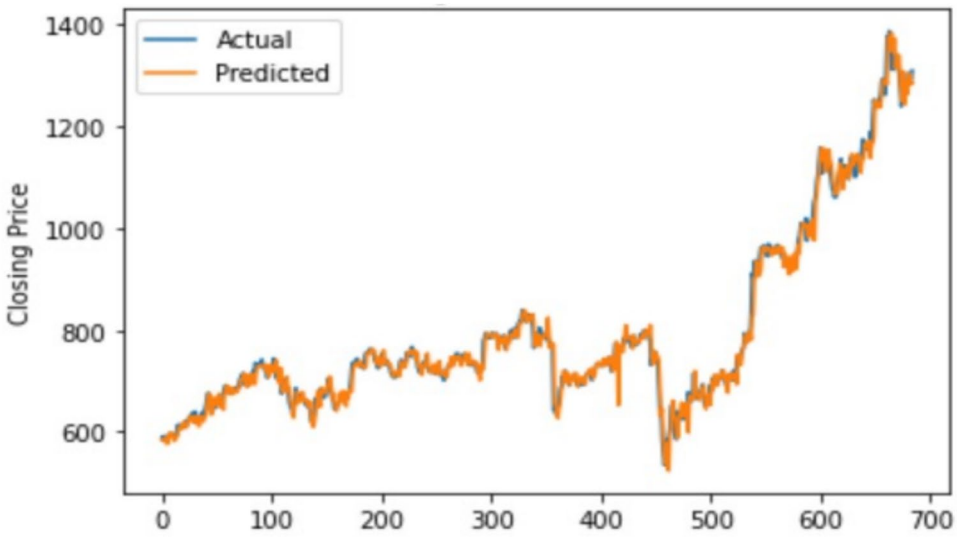


Fig. 13 XGBoost Regression Analysis (Before and During COVID-19 Stock Dataset)

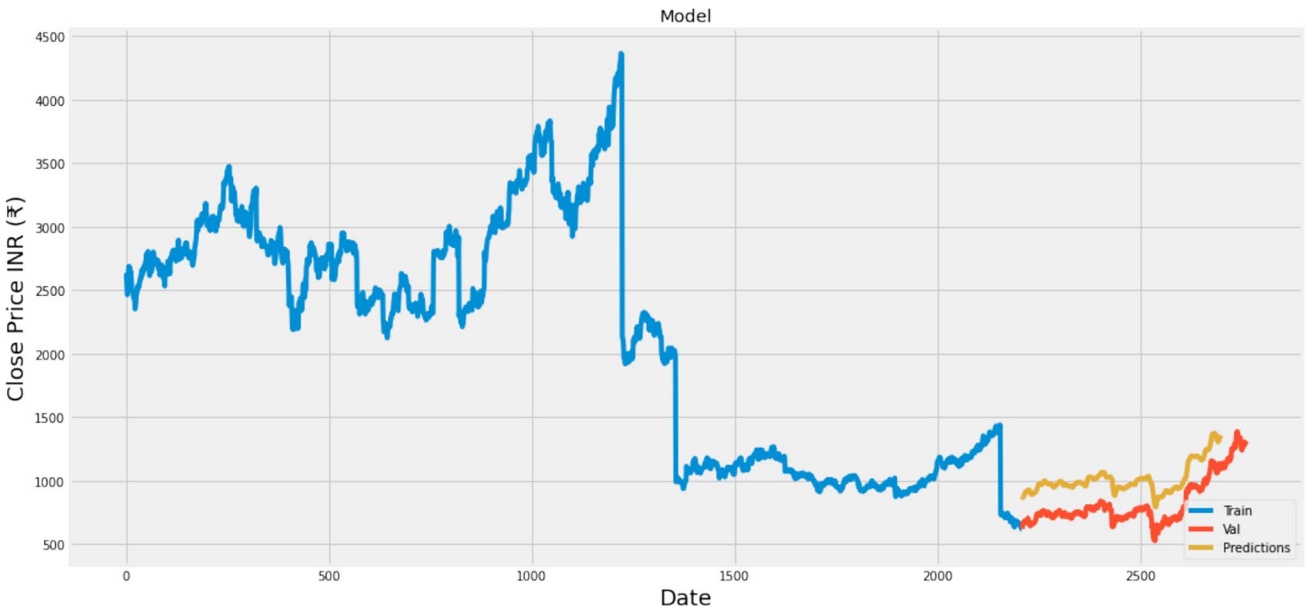


Fig. 14 The output of the LSTM model

Table 9 Metrics of MLP Regressor

Metric	Interpretation	Class 0	Class 1
Precision	Percent of positive instances	0.57	0.62
Recall	Ability to find all positive instances	0.29	0.84
F1-score	The harmonic mean of precision and recall	0.38	0.71
Support	Occurrences of class in a dataset	14	19

reflecting its struggle with abrupt market changes. Similarly, SMA and Holt-Winters showed increased RMSE and MAPE during COVID-19, indicating a limited ability to adapt to extreme volatility. Machine learning models presented diverse outcomes. SVR suffered significant performance degradation, with sharp increases in RMSE, MAE, and MAPE, revealing high sensitivity to market disruptions. RFR and XGBoost demonstrated greater resilience, with XGBoost showing the most stable performance, marked by minimal changes in error metrics. Deep learning models (LSTM and MLP) emerged as the most robust. LSTM maintained consistent RMSE with only a slight increase in MAPE, leveraging its advanced memory mechanisms. MLP delivered the best overall performance, with minimal increases in errors, highlighting its exceptional stability and generalization capability.

	precision	recall	f1-score	support
0	0.49	0.50	0.50	321
1	0.55	0.54	0.55	364
accuracy			0.52	685
macro avg	0.52	0.52	0.52	685
weighted avg	0.52	0.52	0.52	685

Fig. 15 MLP Regressor report

Table 12 compares all models used in this study based on specific parameters. Matrix compares the models used in this study. The accuracy indicates the closeness of the predicted value to the actual stock price. Time-series models are best for short-term forecasting. Long-term prediction means the model can predict stock price trends over a long period. The light-shaded boxes in this category indicate that the proposed model can be used for short-term forecasting. Generalization refers to the ability of a model to adapt to new, unseen data during forecasting. Feature selection is the ability of a model to select the most relevant features for the predicted output.

The results presented in the matrix suggest that deep learning models provide better results than time-series and machine learning models. All time-series models (i.e., ARIMA, SMA, and Holt-Winters) are suitable for short-term predictions. However, these models should not be used for long-term predictions because the error rate increases when stock prices are incorporated into the COVID-19 pandemic data. SVR performed the worst among the machine learning models observed in the study, with the highest error among all applied models. The RFR model had the lowest error among the machine learning models, followed by XGBoost regression. Although the XGBoost and RFR models have higher RMSE than the time series and deep learning models, they are well suited for short-term use and can better predict trend deviation. Therefore, they are more suitable for generalization, that is, more accurate predictions of new unseen data and knowing the trend. MLP had the lowest RMSE, MAE, and MAPE error values among all models in this study. The error values of the MLP were stable for both the pre-COVID-19 and during-COVID-19 datasets, proving its superior performance in predicting the accuracy of long-term forecasting. Although LSTM was more robust to error changes in the pre-COVID-19 and COVID-19 datasets, MLP had better accuracy and very low RMSE, MAE, and MAPE values than LSTM. Among the deep learning models used in this study, MLP is better for stock price prediction. Tables 13 and 14 summarizes the key findings of the model comparisons.

Contributions

Theoretical Contributions

This study makes substantial theoretical contributions to our understanding of financial market prediction mechanisms,

Table 10 Comparison of Market Characteristics and Their Impact on Model Performance: Pre-COVID-19 vs. COVID-19 Periods

Market Characteristics	Pre-COVID-19 Period	COVID-19 Period	Impact on Model Performance
Volatility Characteristics	Stable market conditions with predictable patterns	Extreme volatility with unprecedented price swings	Models struggled to adapt to sudden, erratic market changes due to reliance on pattern continuity
Trading Volume Patterns	Consistent trading volumes	Sporadic volumes with extreme peaks and troughs	Sudden volume fluctuations disrupted models dependent on steady data inputs
Market Sentiment Impact	Driven by traditional market factors	Heavily influenced by global health updates and news	Increased unpredictability in market sentiment reduced model reliability
Price Movement Characteristics	Gradual and steady price trends	Sharp reversals and extreme price swings	Historical-data-based models failed to cope with erratic and abrupt price movements

Table 11 Performance Comparison of Forecasting Models: Pre-COVID-19 vs. COVID-19 Periods

Model Type	Metric	Pre-COVID-19 Value	COVID-19 Value	Performance Insights
Time Series Models				
ARIMA	RMSE	1.32	2.82	More significant absolute errors were made during COVID-19, but relative accuracy improved (MAPE improved: 11.60% → 9.46%)
	MAPE	11.60%	9.46%	
SMA	RMSE	1.3	1.94	Relatively stable, but MAPE increased from 11.46% → 14.49%, indicating prediction accuracy degradation
	MAPE	11.46%	14.49%	
Holt-Winters	RMSE	1.32	2.69	Increased errors during COVID-19; MAPE rose from 11.59% → 13.56%
	MAPE	11.59%	13.56%	
Machine Learning Models				
SVR	RMSE	86.34	249.7	Most severely affected; significant sensitivity to market shifts
	MAE	63.75	164.14	
	MAPE	8.85%	17.48%	
Random Forest Regression	RMSE	14.5	19.57	Relatively stable; minimal increase in MAPE (1.45% → 1.66%) suggests good adaptability
	MAE	9.68	13.15	
	MAPE	1.45%	1.66%	
XGBoost	RMSE	15.75	18.58	Most stable among machine learning models, with minimal changes in MAPE (1.53% → 1.60%)
	MAE	10.26	12.65	
	MAPE	1.53%	1.60%	
Deep Learning Models				
LSTM	RMSE	1.58	1.58	Remarkable stability; slight increase in MAPE (18.23% → 20.02%) due to sophisticated memory mechanisms
	MAE	1.37	1.37	
	MAPE	18.23%	20.02%	
MLP	RMSE	0.6	0.7	Best overall performance and stability, demonstrating robust generalization capabilities
	MAE	0.36	0.5	
	MAPE	0.54%	0.94%	

particularly in how different modeling approaches process and adapt to varying market conditions. It builds upon the foundational work of Mondal et al. [19] and Wen et al. [39] in time-series analysis. Our comparative analysis of time-series, machine learning, and deep learning models reveals important insights that extend beyond mere performance metrics to fundamental theoretical implications for financial forecasting, advancing the systematic framework established by Shah et al. [15].

Central to our theoretical contribution is the advancement of information processing theory in financial markets. Our findings demonstrate that the mechanism by which different models process market information significantly influences their predictive capability, supporting Pai and Lin's [43] observations about information processing constraints. Traditional time-series models, operating on linear information processing principles, showed adequate performance in stable market conditions but demonstrated limitations during market turbulence. This observation challenges the conventional assumption that historical patterns maintain consistent relevance across market conditions [41, 42]. In contrast, deep learning models, particularly LSTM architectures, exhibited remarkable adaptability through their sophisticated non-linear temporal processing capabilities [26, 48, 85], aligning with Zhang et al.'s [38] findings on data-driven prediction systems.

Our analysis provides crucial insights into the theoretical underpinnings of model selection trade-offs in stock price forecasting. SMA and machine learning models, such as RFR and XGBoost, demonstrate reliable short-term and day-to-day forecasting performance, extending Ballings et al.'s [5] evaluation of multiple classifiers. Their ability to capture recent trends and adapt quickly to market changes makes them particularly suitable for immediate decision-making [97]. However, when considering long-term strategic planning and maintaining robust performance across varying

Table 12 Matrix Dashboard

Technique of Forecasting / Parameters of Model Efficiency		Accuracy	Short Term Forecasting	Long Term forecasting	Generalization	Feature Selection
Time Series Models	ARIMA	Medium	High	Low	Low	Low
	SMA	Low	High	Low	Low	Low
	Holt-Winters	High	High	Low	Medium	Medium
Machine Learning Models	SVR	Low	Medium	Low	Low	Low
	RFR	Medium	Medium	Medium	High	High
	XGBoost Regression	Low	Medium	High	High	Low
Deep Learning Models	LSTM	High	Low	High	Low	High
	MLP	High	Low	High	Low	High



market conditions, deep learning models, particularly MLP, prove theoretically superior [98, 99], supporting Long et al.'s [26] research on feature engineering for price movement prediction. These findings contribute significantly to the ongoing discourse on market efficiency theory, building upon Basak et al.'s [23] work on market prediction. The superior performance of machine learning and deep learning models in identifying predictable patterns beyond linear relationships challenges traditional interpretations of the efficient market hypothesis. Rather than supporting a binary view of market efficiency, our results suggest a more nuanced perspective where market efficiency exists on a spectrum that varies with market conditions. This finding aligns with Chandra and He's [33] research on prediction during the COVID-19 pandemic.

Our research extends complexity theory in financial markets by revealing a non-linear relationship between model complexity and prediction accuracy, advancing Nabipour et al.'s [27] work on deep learning for stock market prediction. The inclusion of COVID-19 data provided a unique opportunity to assess the adaptability of various models to significant market shocks, building upon Salisu et al.'s [34] research on financial turbulence. Many conventional approaches struggled to maintain accuracy in extreme volatility and structural changes. At the same time, deep learning models, particularly MLP, showed remarkable resilience, supporting Ren et al.'s [35] findings on extreme risk prediction. A key theoretical advancement emerges from our understanding of feature selection and generalization capabilities in machine learning and deep learning models, extending Peng et al.'s [21] research on feature selection. These advanced techniques demonstrate an impressive ability to identify and leverage the most relevant factors influencing stock prices, supporting Khaidem et al.'s [4] findings on prediction capabilities. This capability proves crucial in complex financial markets, where the relative importance of different factors can rapidly shift, aligning with Na and Kim's [37] work on deep neural networks.

Our analysis also advances model architecture theory by comprehensively examining structural requirements for different market conditions, building upon Rezaei et al.'s [28] research on deep learning applications. We establish theoretical frameworks for understanding how architectural complexity relates to market stability, complementing Kamara et al.'s [29] analysis of ensemble approaches. The trade-offs between model complexity, interpretability, and performance support Ma et al.'s [36] insights on volatility predictability. The importance of model diversity emerges as a crucial theoretical finding in stock price forecasting, extending Lago et al.'s [30] comparative analysis of prediction algorithms. Our research indicates that no single model exists optimally in all scenarios, supporting Siami-Namini and Namin's [31] comparison of traditional and modern approaches. This

Table 13 Summarized of key findings from model comparisons

Model Type	Model	Performance Before COVID-19	Performance During COVID-19	Performance Insights
Time-Series Models	SMA	Reliable for short-term trend detection with low prediction errors	Slightly affected by market volatility, maintaining reasonable performance	Demonstrates resilience in short-term forecasting but struggles with abrupt changes
	ARIMA	Effective in capturing stable historical patterns	Significant challenges in handling sudden and irregular market fluctuations	Relies heavily on pattern continuity, making it less effective during extreme volatility
	Holt-Winters	Suitable for stable trends and seasonality	Struggles with dynamic and volatile conditions	Performs well in structured, predictable environments but is limited in highly erratic markets
Machine Learning Models	SVR	Effective in non-volatile conditions with good accuracy	Severe performance degradation under extreme volatility	Highly sensitive to market shocks, highlighting its unsuitability for volatile conditions
	RFR	Strong generalization with reasonable error rates	Moderately affected by volatility, maintaining fair accuracy	Versatile and adaptable but struggles with sudden market shifts
	XGBoost	Reliable in both stable and moderately volatile conditions	Minimal impact from increased volatility, maintaining robust performance	Best adaptability among machine learning models, suitable for volatile markets
Deep Learning Models	LSTM	Consistently accurate in capturing long-term trends	Maintains robustness and adaptability to volatile market conditions	Excels in handling noise and detecting patterns despite market instability
	MLP	Highly accurate and reliable across conditions	Minimal impact from market volatility; remains stable and precise	Most robust model overall, excelling in both accuracy and adaptability under all conditions

adaptability fundamental aligns with Siami-Namini et al.'s [32] research on forecasting methodologies.

In terms of the method, our study introduces a comprehensive framework for evaluating model performance across different market conditions. This framework extends beyond traditional accuracy metrics to consider stability and adaptability, building upon the work of multiple researchers [30–32] in comparative model analysis. Our findings suggest theoretical foundations for hybrid model development, contributing to the theory of ensemble learning in financial markets.

The integration of these theoretical contributions provides a comprehensive foundation for understanding the relationship between market conditions, model architecture, and prediction performance. Our findings not only advance current theoretical frameworks but also establish new directions for research in financial forecasting and model development, supporting the ongoing evolution of prediction methodologies [97–99]. This theoretical advancement is particularly relevant as financial markets continue to evolve in complexity and interconnectedness, requiring increasingly sophisticated approaches to prediction and analysis while maintaining a balance between accuracy and interpretability.

Practical Contributions

Our findings on stock price forecasting models have significant implications for institutional and individual investors, and regulators in financial ecosystems. The results suggest the need for a diversified approach to financial planning and risk management for institutional investors. Short-term financial planning could benefit from SMA and machine learning models such as RFR and XGBoost, whereas deep learning models, particularly MLP, might better serve long-term strategic planning. The superior performance of MLP in both pre-COVID-19 and COVID-19 datasets indicates its potential for more robust financial forecasting, especially in volatile markets, which could lead to more accurate long-term budgeting and resource allocation. Furthermore, the strong performance of deep learning models suggests that institutional investors should consider investing in AI and machine learning capabilities, potentially by hiring data scientists, upgrading the IT infrastructure, or partnering with fintech firms. These findings on model performance during the COVID-19 pandemic underscore the need for institutional investors to develop flexible forecasting systems that can adapt quickly to unprecedented events. This may involve the creation of ensemble models or systems capable of switching between different forecasting approaches based on market conditions.

The implications for individual investors are substantial, too. Different models can be employed for various aspects of portfolio management, with SMA and machine learning

Table 14 General Observation for Each Forecasting Model

Model Category	General Observation	Key Insights
Time Series Models	Excel in short-term forecasting but lose efficacy in long-term predictions during unprecedented events	Best suited for stable conditions; struggle with abrupt changes and market volatility
Machine Learning Models	Adaptive but volatile; performance degrades with the inclusion of extreme events like COVID-19	Sensitive to sudden market shifts, XGBoost shows the most resilience among machine learning models
Deep Learning Models	Superior capability in long-term forecasting and generalization to unseen data	Most robust and accurate, MLP consistently outperforms others in volatile and stable conditions

models guiding short-term trading decisions, whereas MLP informs long-term investment strategies. The robust performance of the MLP across different market conditions suggests its value in maintaining stable long-term investment strategies, even during market turbulence. The varying performances of the models during the COVID-19 period highlight the importance of using multiple forecasting approaches for comprehensive risk assessment. Individual investors should avoid relying on a single model, particularly during periods of high market volatility. The superior performance of deep learning models, particularly MLP, in terms of accuracy and adaptability to market shocks, indicates their potential application in algorithmic trading systems. However, individual investors must be aware of these models' "black box" nature and potential regulatory implications. While some models better predict short-term trends, investors should exercise caution about over-relying on these for market timing, given the inherent unpredictability of stock markets.

Our study reveals crucial implications for institutional and individual investors in stock price forecasting. A nuanced, multi-model approach is essential, as no single model consistently outperforms across all market conditions. Continuous learning is vital; staying updated on emerging forecasting techniques enables investors to refine strategies and maintain a competitive edge. Awareness of evolving regulatory frameworks surrounding AI and ML in financial decision-making is critical for ensuring compliance and managing risks. Investors must balance innovation with stability, weighing cutting-edge techniques against the need for interpretable strategies. While institutional investors can leverage more complex, resource-intensive approaches, individual investors should focus on accessible tools and understanding limitations. Ultimately, the principles of diversification, continuous improvement, regulatory compliance, and balanced decision-making are crucial for all investors. As the field evolves, maintaining a flexible, informed, and measured approach is key to harnessing the potential of advanced forecasting techniques while effectively managing associated risks.

The findings of this study present several critical considerations for the regulators. The varying performance of the models, especially during the COVID-19 period, emphasizes the need for robust model risk management frameworks. Regulators may require financial institutions to demonstrate the resilience of their forecasting models under various stress scenarios. The superior performance of complex models, such as MLP, raises concerns about transparency and explainability, potentially necessitating new guidelines for using "black-box" models in financial decision-making, and possibly requiring parallel use of more interpretable models.

Regulators should also consider how the widespread use of similar forecasting models might contribute to systemic risk, particularly if many market participants react similarly to the model outputs during stressful events. Given the importance of data quality for model performance, regulators may focus on setting standards for data collection, cleaning, and preprocessing in financial forecasting applications. The rapid evolution of forecasting techniques, as demonstrated by the superior performance of deep learning models, suggests that regulators must continually update their knowledge and guidance, possibly through close collaboration with academic institutions and industry experts. Our findings highlight the need for a nuanced, multi-model approach to stock price forecasting, continuous investment in advanced technologies, and a regulatory framework that balances innovation with stability and transparency. Table 15 presents the practical implications for these three stakeholders.

Conclusion

This study analyzed stock price trends across two datasets from 2011 to 2021, covering pre-COVID-19 and COVID-19 periods. We employed statistical, machine learning, and deep learning models to predict stock prices, addressing the challenges of non-trending or patternless data through advanced techniques such as SVR, XGBoost, and RFR. Deep learning models, including LSTM and MLP, demonstrated superior predictive power, outperforming traditional time-series and machine-learning approaches. Among the models, XGBoost achieved the lowest RMSE, MAE, and MAPE, establishing itself as the most efficient and accurate model in this study.

Table 15 Practical Implications for Institutional and Individual Investors, and Regulators

Stakeholder	Implication	Key Points
Institutional Investors	Risk Management & Financial Planning	Use SMA and ML models (RFR, XGBoost) for short-term; MLP for long-term. MLP's robustness aids in volatile markets
	Investment in AI & ML	Invest in AI/ML capabilities, hire data scientists, upgrade IT, or partner with fintech firms
	Adaptability to Market Shocks	Develop flexible forecasting systems with ensemble models to adapt to market conditions
Individual Investors	Portfolio Management Strategies	Use SMA/ML for short-term trading; MLP for stable long-term investments
	Risk Assessment	Employ multiple models during high market volatility for comprehensive risk assessment
	Algorithmic Trading	Utilize MLP in algorithmic trading but be cautious of the "black box" nature and regulatory implications
	Market Timing	Be cautious of over-relying on short-term predictions for market timing due to inherent unpredictability
Regulators	Model Risk Management	Implement robust frameworks requiring stress scenario resilience demonstrations for forecasting models
	Transparency & Explainability	Develop guidelines for "black box" models, possibly requiring parallel interpretable models
	Systemic Risk Assessment	Consider systemic risk from widespread use of similar models, especially during market shocks
	Data Quality & Standardization	Set standards for data collection, cleaning, and preprocessing in forecasting applications
	Continuing Education & Guidance	Update regulatory knowledge and guidance in collaboration with academia and industry experts

Unlike previous research that relied heavily on serial model predictions influenced by parameters like time intervals and market instruments, this study precisely maps parameters to guide model selection. Our findings highlight the contextual strengths of different models: time-series models excel in short-term forecasts, while deep learning models like LSTM demonstrate enhanced accuracy across both short- and long-term horizons. Machine learning models, particularly RFR, offer robust generalization and adaptability.

Future research should refine financial market prediction by expanding XGBoost applications across diverse instruments such as stocks, currencies, and commodities. Integrating advanced feature engineering and incorporating novel data sources—like social media trends and macroeconomic indicators—can further enhance predictive frameworks. Additionally, optimizing XGBoost hyperparameters and synergizing it with deep learning architectures, such as LSTM and transformer models, could unlock new possibilities for superior market forecasting.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. This study did not receive any funding.

Declarations

Conflict of Interest All Authors (Mugdha Shailendra Kulkarni, S. Vijayakumar Bharathi, Arif Perdana, and Divisha Kilari) declare that he/she has no conflict of interest.

Ethical approval Ethical approval is not applicable because this article contains no studies with human participants or animals performed by any of the authors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Friberg R, Goldstein I, Hankins KW. Corporate responses to stock price fragility. J Financ Econ. 2024;153: 103795. https://doi.org/10.1016/j.jfineco.2024.103795.
2. Kumbure MM, Lohrmann C, Luukka P, Porras J. Machine learning techniques and data for stock market forecasting: a literature

review. Expert Syst Appl. 2022;197: 116659. https://doi.org/10.1016/j.eswa.2022.116659.
3. Jagwani J, Gupta M, Sachdeva H, Singhal A. Stock price forecasting using data from yahoo finance and analysing seasonal and nonseasonal trend. 2018 Second International Conference on Intelligent Computing and Control Systems (ICICCS). IEEE; 2018. pp. 462–7. https://doi.org/10.1109/ICCONS.2018.8663035.
4. Khaidem L, Saha S, Dey SR. Predicting the direction of stock market prices using random forest. 2016. https://doi.org/10.48550/arXiv.1605.00003
5. Ballings M, Van den Poel D, Hespeels N, Gryp R. Evaluating multiple classifiers for stock price direction prediction. Expert Syst Appl. 2015;42:7046–56. https://doi.org/10.1016/j.eswa.2015.05.013.
6. Dhingra K, Kapil S. Impact of macroeconomic variables on stock market—an empirical study. In: Lakhanpal P, Mukherjee J, Nag B, Tuteja D, editors. Trade, investment and economic growth. Singapore: Springer Singapore; 2021. p. 177–94. https://doi.org/10.1007/978-981-33-6973-3_12.
7. Ouattara BS. Using macroeconomic variables in the prediction of stock market indices: A theoretical and empirical assessment within BRICS and selected developed economies. London South Bank University, 2018.
8. Malikov K, Demirbag M, Kuvandikov A, Manson S. Workforce reductions and post-merger operating performance: the role of corporate governance. J Bus Res. 2021;122:109–20. https://doi.org/10.1016/j.jbusres.2020.08.042.
9. Hao P-Y, Kung C-F, Chang C-Y, Ou J-B. Predicting stock price trends based on financial news articles and using a novel twin support vector machine with fuzzy hyperplane. Appl Soft Comput. 2021;98: 106806. https://doi.org/10.1016/j.asoc.2020.106806.
10. Li X, Wu P, Wang W. Incorporating stock prices and news sentiments for stock market prediction: a case of Hong Kong. Inf Process Manage. 2020;57: 102212. https://doi.org/10.1016/j.ipm.2020.102212.
11. Vanstone BJ, Gepp A, Harris G. Do news and sentiment play a role in stock price prediction? Appl Intell. 2019;49:3815–20. https://doi.org/10.1007/s10489-019-01458-9.
12. Bikhchandani S, Sharma S. Herd behavior in financial markets. IMF Staff Pap. 2000;47(3):279–310. https://doi.org/10.2307/3867650.
13. Nofsinger JR. The psychology of investing. London: Routledge; 2017.
14. Picasso A, Merello S, Ma Y, Oneto L, Cambria E. Technical analysis and sentiment embeddings for market trend prediction. Expert Syst Appl. 2019;135:60–70. https://doi.org/10.1016/j.eswa.2019.06.014.
15. Shah D, Isah H, Zulkernine F. Stock market analysis: a review and taxonomy of prediction techniques. Int J Financ Stud. 2019;7:26. https://doi.org/10.3390/ijfs7020026.
16. Bisoi R, Dash PK, Parida AK. Hybrid variational mode decomposition and evolutionary robust kernel extreme learning machine for stock price and movement prediction on daily basis. Appl Soft Comput. 2019;74:652–78. https://doi.org/10.1016/j.asoc.2018.11.008.
17. Zhang D, Lou S. The application research of neural network and BP algorithm in stock price pattern classification and prediction. Futur Gener Comput Syst. 2021;115:872–9. https://doi.org/10.1016/j.future.2020.10.009.
18. Atsalakis GS, Valavanis KP. Surveying stock market forecasting techniques—Part II: soft computing methods. Expert Syst Appl. 2009;36:5932–41. https://doi.org/10.1016/j.eswa.2008.07.006.
19. Mondal P, Shit L, Goswami S. Study of effectiveness of time series modeling (Arima) in forecasting stock prices. Int J Comput Sci Eng Appl. 2014;4:13–29. https://doi.org/10.5121/ijcsea.2014.4202.
20. Mottaghi N, Farhangdoost S. Stock price forecasting in presence of covid-19 pandemic and evaluating performances of machine learning models for time-series forecasting. 2021.
21. Peng Y, Albuquerque PHM, Kimura H, Saavedra CAPB. Feature selection and deep neural networks for stock price direction forecasting using technical analysis indicators. Mach Learn Appl. 2021;5: 100060. https://doi.org/10.1016/j.mlwa.2021.100060.
22. Nazareth N, Ramana Reddy YV. Financial applications of machine learning: a literature review. Expert Syst Appl. 2023;219: 119640. https://doi.org/10.1016/j.eswa.2023.119640.
23. Basak S, Kar S, Saha S, Khaidem L, Dey SR. Predicting the direction of stock market prices using tree-based classifiers. North Am J Econ Finan. 2019;47:552–67. https://doi.org/10.1016/j.najef.2018.06.013.
24. Chong E, Han C, Park FC. Deep learning networks for stock market analysis and prediction: methodology, data representations, and case studies. Expert Syst Appl. 2017;83:187–205. https://doi.org/10.1016/j.eswa.2017.04.030.
25. Long J, Chen Z, He W, Wu T, Ren J. An integrated framework of deep learning and knowledge graph for prediction of stock price trend: An application in Chinese stock exchange market. Appl Soft Comput. 2020;91: 106205. https://doi.org/10.1016/j.asoc.2020.106205.
26. Long W, Lu Z, Cui L. Deep learning-based feature engineering for stock price movement prediction. Knowl-Based Syst. 2019;164:163–73. https://doi.org/10.1016/j.knosys.2018.10.034.
27. Nabipour M, Nayyeri P, Jabani H, Mosavi A, Salwana ES. Deep learning for stock market prediction. Entropy. 2020;22:840. https://doi.org/10.3390/e22080840.
28. Rezaei H, Faaljou H, Mansourfar G. Stock price prediction using deep learning and frequency decomposition. Expert Syst Appl. 2021;169: 114332. https://doi.org/10.1016/j.eswa.2020.114332.
29. Kamara AF, Chen E, Pan Z. An ensemble of a boosted hybrid of deep learning models and technical analysis for forecasting stock prices. Inf Sci. 2022;594:1–19. https://doi.org/10.1016/j.ins.2022.02.015.
30. Lago J, De Ridder F, De Schutter B. Forecasting spot electricity prices: deep learning approaches and empirical comparison of traditional algorithms. Appl Energy. 2018;221:386–405. https://doi.org/10.1016/j.apenergy.2018.02.069.
31. Siami-Namini S, Siami-Namin A. Forecasting economics and financial time series: ARIMA vs. LSTM. 2018. https://doi.org/10.48550/arXiv.1803.06386
32. Siami-Namini S, Namin AS. Forecasting economics and financial time series: ARIMA vs. LSTM. 2018. https://doi.org/10.48550/ARXIV.1803.06386.
33. Chandra R, He Y. Bayesian neural networks for stock price forecasting before and during COVID-19 pandemic. PLoS ONE. 2021;16: e0253217. https://doi.org/10.1371/journal.pone.0253217.
34. Salisu AA, Demirer R, Gupta R. Financial turbulence, systemic risk and the predictability of stock market volatility. Glob Financ J. 2022;52: 100699. https://doi.org/10.1016/j.gfj.2022.100699.
35. Ren T, Li S, Zhang S. Stock market extreme risk prediction based on machine learning: evidence from the American market. North Am J Econ Finan. 2024;74: 102241. https://doi.org/10.1016/j.najef.2024.102241.
36. Ma F, Wang J, Wahab MIM, Ma Y. Stock market volatility predictability in a data-rich world: a new insight. Int J Forecast. 2023;39:1804–19. https://doi.org/10.1016/j.ijforecast.2022.08.010.
37. Na H, Kim S. Predicting stock prices based on informed traders' activities using deep neural networks. Econ Lett. 2021;204: 109917. https://doi.org/10.1016/j.econlet.2021.109917.

38. Zhang J, Cui S, Xu Y, Li Q, Li T. A novel data-driven stock price trend prediction system. Expert Syst Appl. 2018;97:60–9. https://doi.org/10.1016/j.eswa.2017.12.026.
39. Wen M, Li P, Zhang L, Chen Y. Stock market trend prediction using high-order information of time series. IEEE Access. 2019;7:28299–308. https://doi.org/10.1109/ACCESS.2019.2901842.
40. Wu J, Xu K, Chen X, Li S, Zhao J. Price graphs: utilizing the structural information of financial time series for stock prediction. Inf Sci. 2022;588:405–24. https://doi.org/10.1016/j.ins.2021.12.089.
41. Ariyo AA, Adewumi AO, Ayo CK. Stock Price Prediction Using the ARIMA Model. 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation. IEEE; 2014. pp. 106–12. https://doi.org/10.1109/UKSim.2014.67.
42. Phuoc T, Anh PTK, Tam PH, Nguyen CV. Applying machine learning algorithms to predict the stock price trend in the stock market—the case of Vietnam. Humanit Soc Sci Commun. 2024;11:393. https://doi.org/10.1057/s41599-024-02807-x.
43. Pai P-F, Lin C-S. A hybrid ARIMA and support vector machines model in stock price forecasting. Omega. 2005;33:497–505. https://doi.org/10.1016/j.omega.2004.07.024.
44. Nti IK, Adekoya AF, Weyori BA. A systematic review of fundamental and technical analysis of stock market predictions. Artif Intell Rev. 2020;53:3007–57. https://doi.org/10.1007/s10462-019-09754-z.
45. Sheth D, Shah M. Predicting stock market using machine learning: best and accurate way to know future stock prices. Int J Syst Assur Eng Manag. 2023;14:1–18. https://doi.org/10.1007/s13198-022-01811-1.
46. Lu C-J, Lee T-S, Chiu C-C. Financial time series forecasting using independent component analysis and support vector regression. Decis Support Syst. 2009;47:115–25. https://doi.org/10.1016/j.dss.2009.02.001.
47. Illa PK, Parvathala B, Sharma AK. Stock price prediction methodology using random forest algorithm and support vector machine. Mater Today Proc. 2021. https://doi.org/10.1016/j.matpr.2021.10.460.
48. Li Z, Tam V. A comparative study of a recurrent neural network and support vector machine for predicting price movements of stocks of different volatilites. 2017 IEEE Symposium Series on Computational Intelligence (SSCI). IEEE; 2017. pp. 1–8. https://doi.org/10.1109/SSCI.2017.8285319.
49. Hafezi R, Shahrabi J, Hadavandi E. A bat-neural network multi-agent system (BNNMAS) for stock price prediction: case study of DAX stock price. Appl Soft Comput. 2015;29:196–210. https://doi.org/10.1016/j.asoc.2014.12.028.
50. Subasi A, Erçelebi E. Classification of EEG signals using neural network and logistic regression. Comput Methods Progr Biomed. 2005;78:87–99. https://doi.org/10.1016/j.cmpb.2004.10.009.
51. Mat Isa NA, Mamat WMFW. Clustered-hybrid multilayer perceptron network for pattern recognition application. Appl Soft Comput. 2011;11:1457–66. https://doi.org/10.1016/j.asoc.2010.04.017.
52. Klein JL, Klein D. Statistical visions in time: a history of time series analysis. Cambridge: Cambridge University Press; 1997.
53. Shi H-L, Zhou W-X. Time series momentum and contrarian effects in the Chinese stock market. Phys A. 2017;483:309–18. https://doi.org/10.1016/j.physa.2017.04.139.
54. Spiro AG, Goldovskaya MD, Kiseleva NE, Pokrovskaya IV. Segmentation and hashing of time series in stock market prediction. Autom Remote Control. 2018;79:911–8. https://doi.org/10.1134/S0005117918050119.
55. Hamilton JD. Time series analysis. Princeton: Princeton University Press; 1994.
56. Wang J-H, Leu J-Y. Stock market trend prediction using ARIMA-based neural networks. Proceedings of International Conference on Neural Networks (ICNN'96). IEEE; 1996. pp. 2160–5. https://doi.org/10.1109/ICNN.1996.549236.
57. Mills TC. Applied time series analysis: a practical guide to modeling and forecasting. London: Academic Press; 2019.
58. Hossain E, Hossain MS, Zander P-O, Andersson K. Machine learning with belief rule-based expert systems to predict stock price movements. Expert Syst Appl. 2022;206: 117706. https://doi.org/10.1016/j.eswa.2022.117706.
59. Oriani FB, Coelho GP. Evaluating the impact of technical indicators on stock forecasting. 2016 IEEE Symposium Series on Computational Intelligence (SSCI). IEEE; 2016. pp. 1–8. https://doi.org/10.1109/SSCI.2016.7850017.
60. Billah MM, Sultana A, Bhuiyan F, Kaosar MG. Stock price prediction: comparison of different moving average techniques using deep learning model. Neural Comput Appl. 2024;36:5861–71. https://doi.org/10.1007/s00521-023-09369-0.
61. Svetunkov I, Petropoulos F. Old dog, new tricks: a modelling view of simple moving averages. Int J Prod Res. 2018;56:6034–47. https://doi.org/10.1080/00207543.2017.1380326.
62. Licona-Luque JP, Brenes-García LF, Cantú-Ortiz FJ, Ceballos-Cancino HG. Simple Moving Average (SMA) Investment Strategy During COVID-19 Pandemic. In: Yang X-S, Sherratt RS, Dey N, Joshi A, editors. Proceedings of Eighth International Congress on Information and Communication Technology, vol. 694. Singapore: Springer Nature Singapore; 2023. pp. 545–55. https://doi.org/10.1007/978-981-99-3091-3_45.
63. Mak DK. Simple moving average. Trading tactics in the financial market. Cham: Springer International Publishing; 2021. pp. 29–55. https://doi.org/10.1007/978-3-030-70622-7_3.
64. Wang Y, Guo Y. Forecasting method of stock market volatility in time series data based on mixed model of ARIMA and XGBoost. China Commun. 2020;17:205–21. https://doi.org/10.23919/JCC.2020.03.017.
65. Verma S, Prakash Sahu S, Prasad Sahu T. Ensemble Approach for Stock Market Forecasting Using ARIMA and LSTM Model. In: Pandian AP, Palanisamy R, Narayanan M, Senjyu T, editors. Proceedings of Third International Conference on Intelligent Computing, Information and Control Systems, vol. 1415. Singapore: Springer Nature Singapore; 2022. pp. 65–80. https://doi.org/10.1007/978-981-16-7330-6_6.
66. Stoean R, Stoean C, Sandita A. Evolutionary Regressor Selection in ARIMA Model for Stock Price Time Series Forecasting. In: Czarnowski I, Howlett RJ, Jain LC, editors. Intelligent Decision Technologies 2017, vol. 73. Cham: Springer International Publishing; 2018. pp. 117–26. https://doi.org/10.1007/978-3-319-59424-8_11.
67. Waheed W, Qingshan X. An efficient load forecasting technique by using Holt-Winters and Prophet algorithms to mitigate the impact on power consumption in COVID-19. IET Energy Syst Integration. 2024;esi2.12132. https://doi.org/10.1049/esi2.12132.
68. Rumbe G, Hamasha M, Al MS. A comparison of Holts-Winter and artificial neural network approach in forecasting: a case study for tent manufacturing industry. Res Eng. 2024;21: 101899. https://doi.org/10.1016/j.rineng.2024.101899.
69. Behera G, Bhoi AK, Bhoi A. UHWSF: Univariate holt winter's based store sales forecasting. In: Udgata SK, Sethi S, Srirama SN (eds) Intelligent systems, vol. 185. Singapore: Springer Singapore; 2021. pp. 283–92. https://doi.org/10.1007/978-981-33-6081-5_25.
70. Tiwari S, Bharadwaj A, Gupta S. Stock price prediction using data analytics. 2017 International Conference on Advances in Computing, Communication and Control (ICAC3). IEEE; 2017. pp. 1–5. https://doi.org/10.1109/ICAC3.2017.8318783.

71. Chatfield C. The holt-winters forecasting procedure. Appl Stat. 1978;27:264. https://doi.org/10.2307/2347162.
72. Gelper S, Fried R, Croux C. Robust forecasting with exponential and Holt-Winters smoothing. J Forecast 2009;n/a-n/a. https://doi.org/10.1002/for.1125.
73. Ansari GA, Bhat SS, Ansari MD, Ahmad S, Nazeer J, Eljialy AEM. Performance evaluation of machine learning techniques (MLT) for heart disease prediction. Comput Math Methods Med. 2023;2023:8191261. https://doi.org/10.1155/2023/8191261.
74. Tsai C-F, Lin Y-C, Yen DC, Chen Y-M. Predicting stock returns by classifier ensembles. Appl Soft Comput. 2011;11:2452–9. https://doi.org/10.1016/j.asoc.2010.10.001.
75. Henrique BM, Sobreiro VA, Kimura H. Stock price prediction using support vector regression on daily and up to the minute prices. J Finan Data Sci. 2018;4:183–201. https://doi.org/10.1016/j.jfds.2018.04.003.
76. Lahmiri S. Minute-ahead stock price forecasting based on singular spectrum analysis and support vector regression. Appl Math Comput. 2018;320:444–51. https://doi.org/10.1016/j.amc.2017.09.049.
77. Mishra S, Padhy S. An efficient portfolio construction model using stock price predicted by support vector regression. North Am J Econ Finan. 2019;50: 101027. https://doi.org/10.1016/j.najef.2019.101027.
78. Hsu S-H, Hsieh JP-A, Chih T-C, Hsu K-C. A two-stage architecture for stock price forecasting by integrating self-organizing map and support vector regression. Expert Syst Appl 2009;36:7947–51. https://doi.org/10.1016/j.eswa.2008.10.065.
79. Kara Y, Acar Boyacioglu M, Baykan ÖK. Predicting direction of stock price index movement using artificial neural networks and support vector machines: the sample of the Istanbul stock exchange. Expert Syst Appl. 2011;38:5311–9. https://doi.org/10.1016/j.eswa.2010.10.027.
80. Breiman L. Random forests. Mach Learn. 2001;45:5–32. https://doi.org/10.1023/A:1010933404324.
81. Weng B, Lu L, Wang X, Megahed FM, Martinez W. Predicting short-term stock prices using ensemble methods and online data sources. Expert Syst Appl. 2018;112:258–73. https://doi.org/10.1016/j.eswa.2018.06.016.
82. Park HJ, Kim Y, Kim HY. Stock market forecasting using a multi-task approach integrating long short-term memory and the random forest framework. Appl Soft Comput. 2022;114: 108106. https://doi.org/10.1016/j.asoc.2021.108106.
83. Yuan X, Yuan J, Jiang T, Ain QU. Integrated long-term stock selection models based on feature selection and machine learning algorithms for china stock market. IEEE Access. 2020;8:22672–85. https://doi.org/10.1109/ACCESS.2020.2969293.
84. Liu J, Zhang S, Fan H. A two-stage hybrid credit risk prediction model based on XGBoost and graph-based deep neural network. Expert Syst Appl. 2022;195: 116624. https://doi.org/10.1016/j.eswa.2022.116624.
85. Fischer T, Krauss C. Deep learning with long short-term memory networks for financial market predictions. Eur J Oper Res. 2018;270:654–69. https://doi.org/10.1016/j.ejor.2017.11.054.
86. Bhoite S, Ansari G, Patil CH, Thatte S, Magar V, Gandhi K. Stock Market Prediction Using Recurrent Neural Network and Long Short-Term Memory. In: Tuba M, Akashe S, Joshi A, editors. ICT Infrastructure and Computing, vol. 520. Singapore: Springer Nature Singapore; 2023. pp. 635–43. https://doi.org/10.1007/978-981-19-5331-6_65.
87. Sezer OB, Gudelek MU, Ozbayoglu AM. Financial time series forecasting with deep learning : a systematic literature review: 2005–2019. Appl Soft Comput. 2020;90: 106181. https://doi.org/10.1016/j.asoc.2020.106181.
88. Liu K, Zhou J, Dong D. Improving stock price prediction using the long short-term memory model combined with online social networks. J Behav Exp Financ. 2021;30: 100507. https://doi.org/10.1016/j.jbef.2021.100507.
89. Shah D, Campbell W, Zulkernine FH. A Comparative Study of LSTM and DNN for Stock Market Forecasting. 2018 IEEE International Conference on Big Data (Big Data). IEEE; 2018. pp. 4148–55. https://doi.org/10.1109/BigData.2018.8622462.
90. Sujatha KV, Sundaram SM. Stock index prediction using regression and neural network models under non normal conditions. INTERACT-2010. IEEE; 2010. pp. 59–63. https://doi.org/10.1109/INTERACT.2010.5706195.
91. Namdari A, Durrani TS. A multilayer feedforward perceptron model in neural networks for predicting stock market short-term trends. SN Oper Res Forum. 2021;2:38. https://doi.org/10.1007/s43069-021-00071-2.
92. Chandra J, Nachamai M, Pillai AS. Empirical Estimation of Multilayer Perceptron for Stock Market Indexes. In: Sulaiman HA, Othman MA, Othman MFI, Rahim YA, Pee NC, editors. Advanced Computer and Communication Engineering Technology, vol. 362. Cham: Springer International Publishing; 2016. pp. 747–58. https://doi.org/10.1007/978-3-319-24584-3_63.
93. Lantz B. Machine learning with R: expert techniques for predictive modelling. Ltd.: Packt Publishing; 2019.
94. Tan P-N, Steinbach M, Kumar V, Karpatne A. Introduction to data mining, 2nd Edition. Pearson; 2019.
95. Breiman L. Bagging predictors. Mach Learn. 1996;24:123–40. https://doi.org/10.1007/BF00058655.
96. Chai T, Draxler RR. Root mean square error (RMSE) or mean absolute error (MAE)?—Arguments against avoiding RMSE in the literature. Geosci Model Dev. 2014;7:1247–50. https://doi.org/10.5194/gmd-7-1247-2014.
97. Gokce MM, Duman E. Performance Comparison of Simple Regression, Random Forest and XGBoost Algorithms for Forecasting Electricity Demand. 2022 3rd International Informatics and Software Engineering Conference (IISEC). Ankara, Turkey: IEEE; 2022. pp. 1–6. https://doi.org/10.1109/IISEC56263.2022.9998213.
98. Bian N, Zhu M, Chen L, Cai W. Boosting MLPs with a coarsening strategy for long-term time series forecasting. 2024. https://doi.org/10.48550/ARXIV.2405.03199.
99. Bachmann G, Anagnostidis S, Hofmann T. Scaling MLPs: a tale of inductive bias. 2023. https://doi.org/10.48550/ARXIV.2306.13575.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.